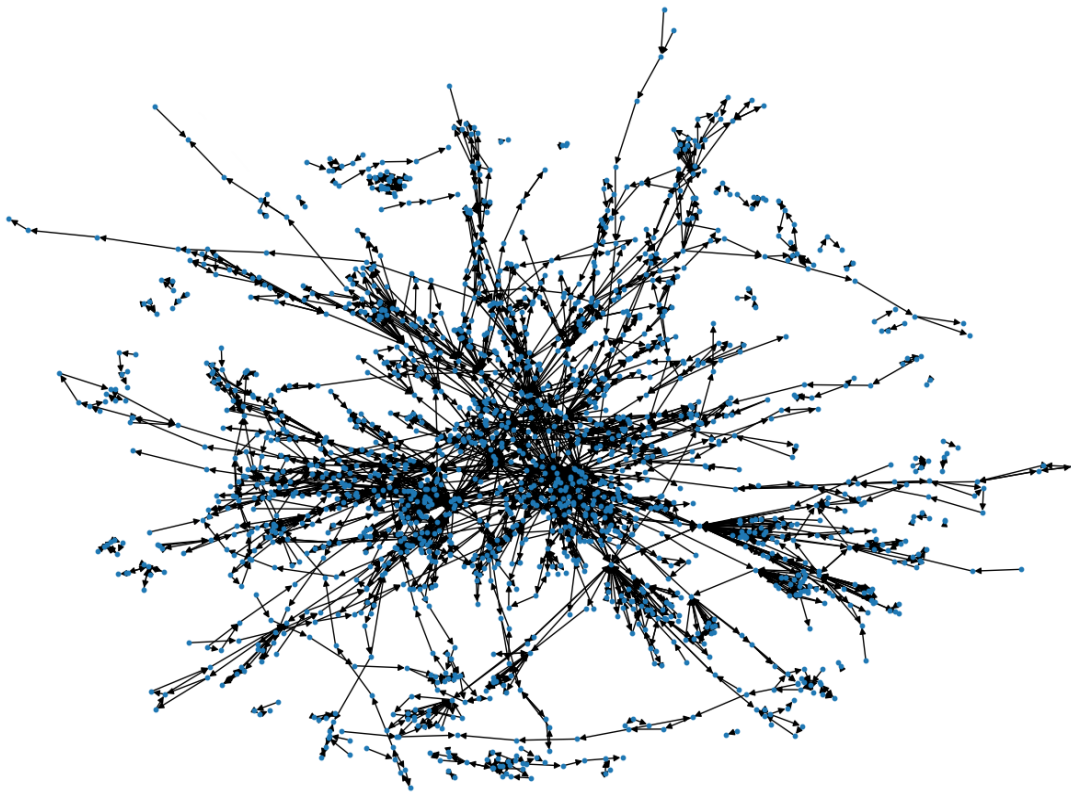


Network Operations Strategy Memo

This memo presents a graph-driven logistics network intelligence framework designed to identify bottleneck hubs, high-risk corridors, and ETA instability patterns across shipment flows. The analysis combines graph analytics, machine learning-based ETA prediction, and operational strategy modeling to recommend actionable logistics interventions.

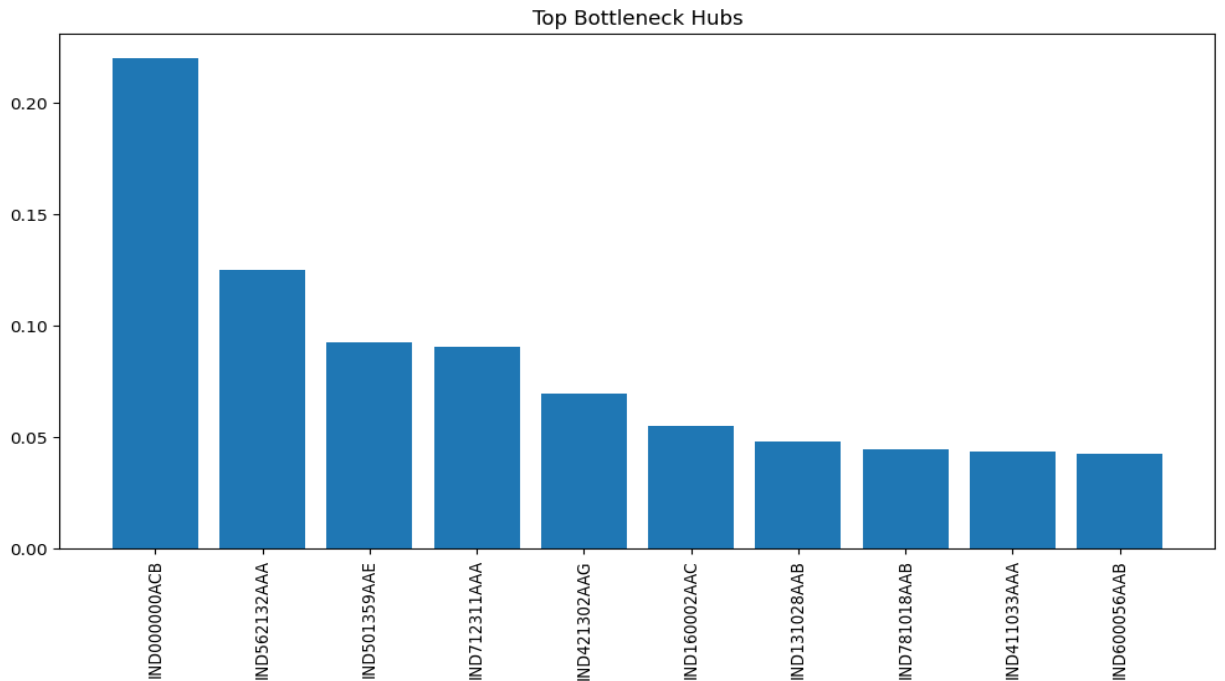
1. Logistics Network Graph Structure

The logistics system was modeled as a directed weighted graph where: Nodes represent logistics hubs or shipment centers. Edges represent shipment movement corridors. Edge weights capture delay ratios and corridor intensity. The graph reveals strong central hub dependencies and corridor clustering patterns. A few highly connected hubs dominate shipment movement, making them operationally critical and highly vulnerable to congestion-related cascading delays.



2. Top Bottleneck Hubs

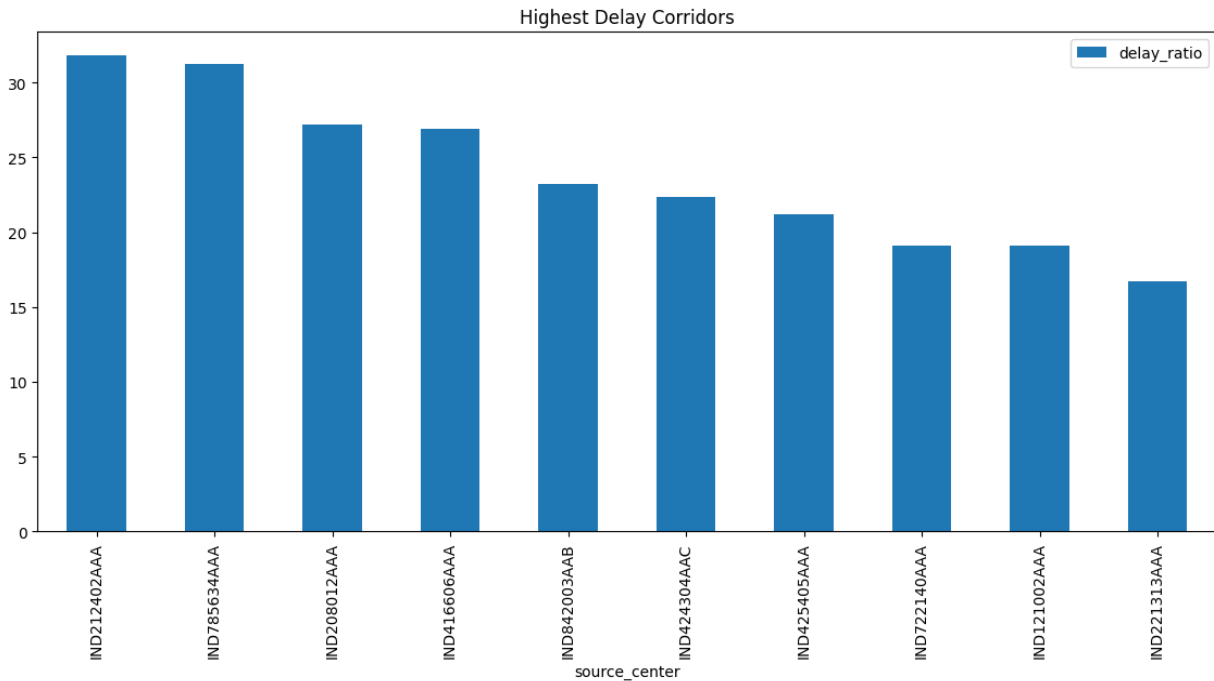
Betweenness centrality analysis identified several critical hubs acting as dominant transit dependencies. These hubs carry disproportionately high shipment flow and significantly impact overall network resilience.



Hub	Risk Pattern	Operational Impact	Recommended Intervention
IND00000ACB	Extreme network dependency	Cascading delay amplification	Dedicated FTL prioritization
IND562132AAA	High corridor convergence	Hub congestion	Dynamic route balancing
IND501359AAE	Multi-hop dependency	Transit backlog risk	Micro-hub distribution
IND712311AAA	Heavy shipment inflow	ETA instability	Priority shipment segregation
IND421302AAG	Corridor overload	SLA breach exposure	Additional fleet deployment

3. Highest Delay Corridors

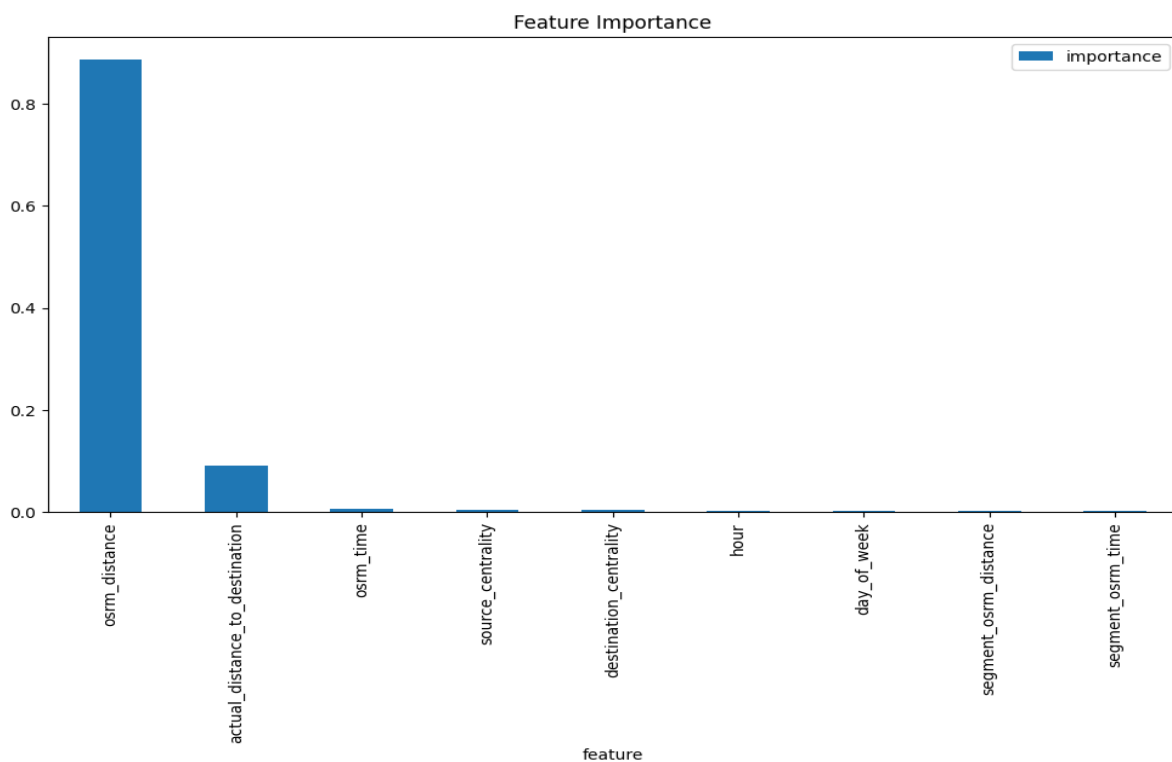
Delay-ratio analysis identified several corridors exhibiting structurally high transit inefficiencies. These corridors are characterized by excessive transit variability, operational fragmentation, and unstable ETA behavior.



Key Recommendations: Shift high-priority shipments on unstable corridors from Carting to FTL operations Add dynamic dispatch buffers for high-variance corridors Introduce alternate routing for corridors with repeated SLA failures Deploy corridor-level monitoring dashboards for live intervention

4. ETA Prediction Insights

The ETA prediction model shows that OSRM distance overwhelmingly drives shipment transit prediction, followed by actual-distance ratios and network-centrality features. This indicates that physical network structure and corridor topology strongly influence logistics reliability.



5. FTL vs Carting Decision Framework

The operational strategy recommends transportation mode allocation using corridor intelligence and shipment risk scoring. **FTL Recommended For:** High-value shipments Time-sensitive enterprise deliveries Stable high-volume corridors Corridors with excessive handling dependencies **Carting Recommended For:** Low-density shipment corridors Non-critical deliveries Flexible cost-sensitive operations Rural fragmented distribution networks

Parameter	FTL Preferred	Carting Preferred
Shipment Volume	High	Low
Delay Variance	Low	Moderate
SLA Criticality	High	Medium
Transit Stability	Stable	Flexible
Handling Dependency	Low	High
Delivery Urgency	High	Low

6. Revenue Impact Estimates

The proposed operational interventions are expected to generate measurable business impact: 8–12% reduction in SLA penalty exposure 10–18% improvement in operational efficiency 5–10% increase in effective corridor throughput Improved fleet utilization and dispatch planning Higher customer retention through improved ETA reliability Strategic corridor optimization combined with selective FTL deployment can substantially reduce cascading delays and improve overall network resilience.

Conclusion

This project demonstrates how graph analytics, machine learning, and corridor intelligence can transform logistics operations into a scalable, data-driven network optimization system. The integration of bottleneck analysis, ETA prediction, and transportation strategy recommendations enables smarter logistics planning and measurable operational impact.